ASSIGNMENT 1

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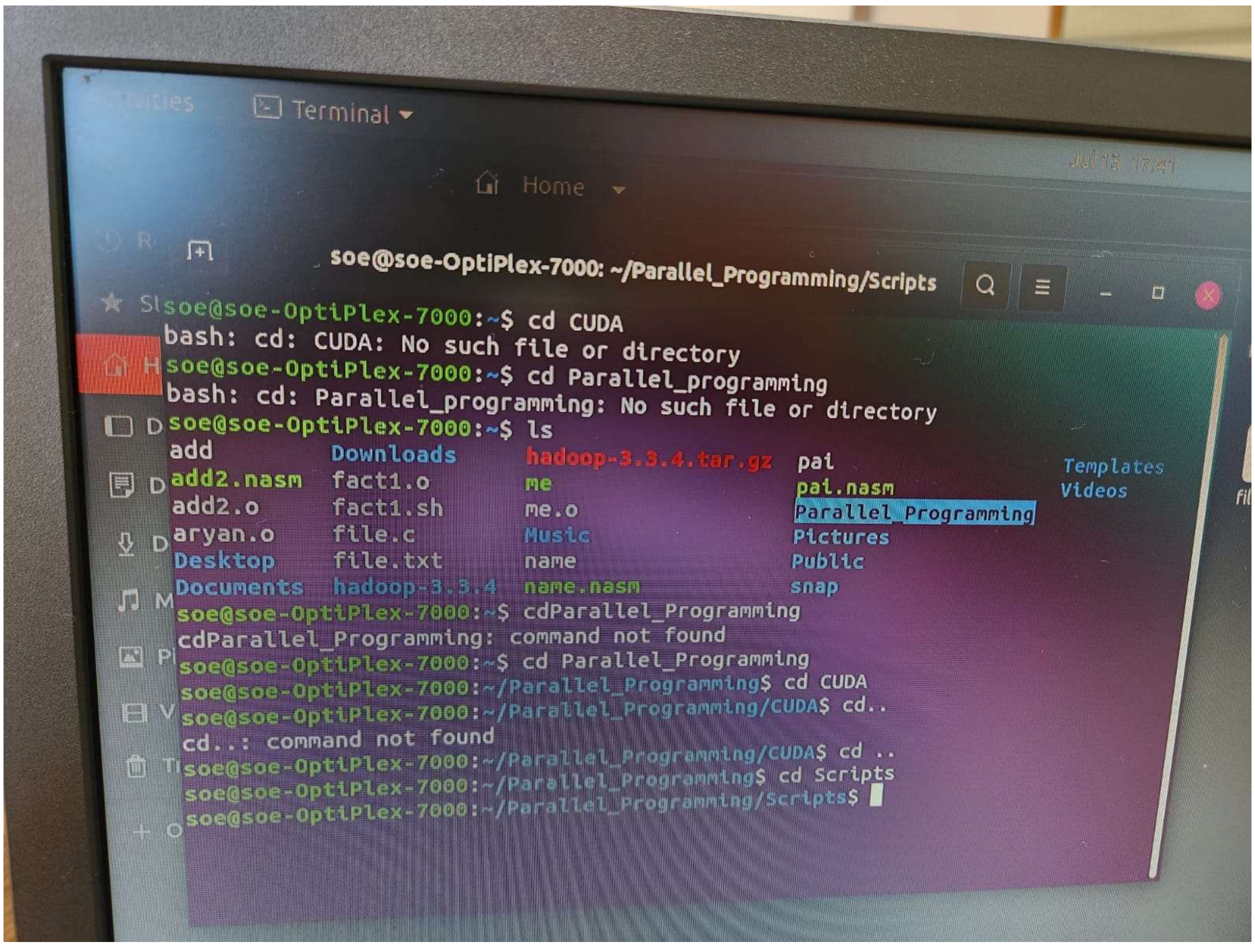
SUB: HPCL

Q1. Directory Creation

Task: Create the following directory structure inside the home folder:
Parallel_Programming/ with subdirectories OpenMP/, MPI/, CUDA/, and Scripts/.



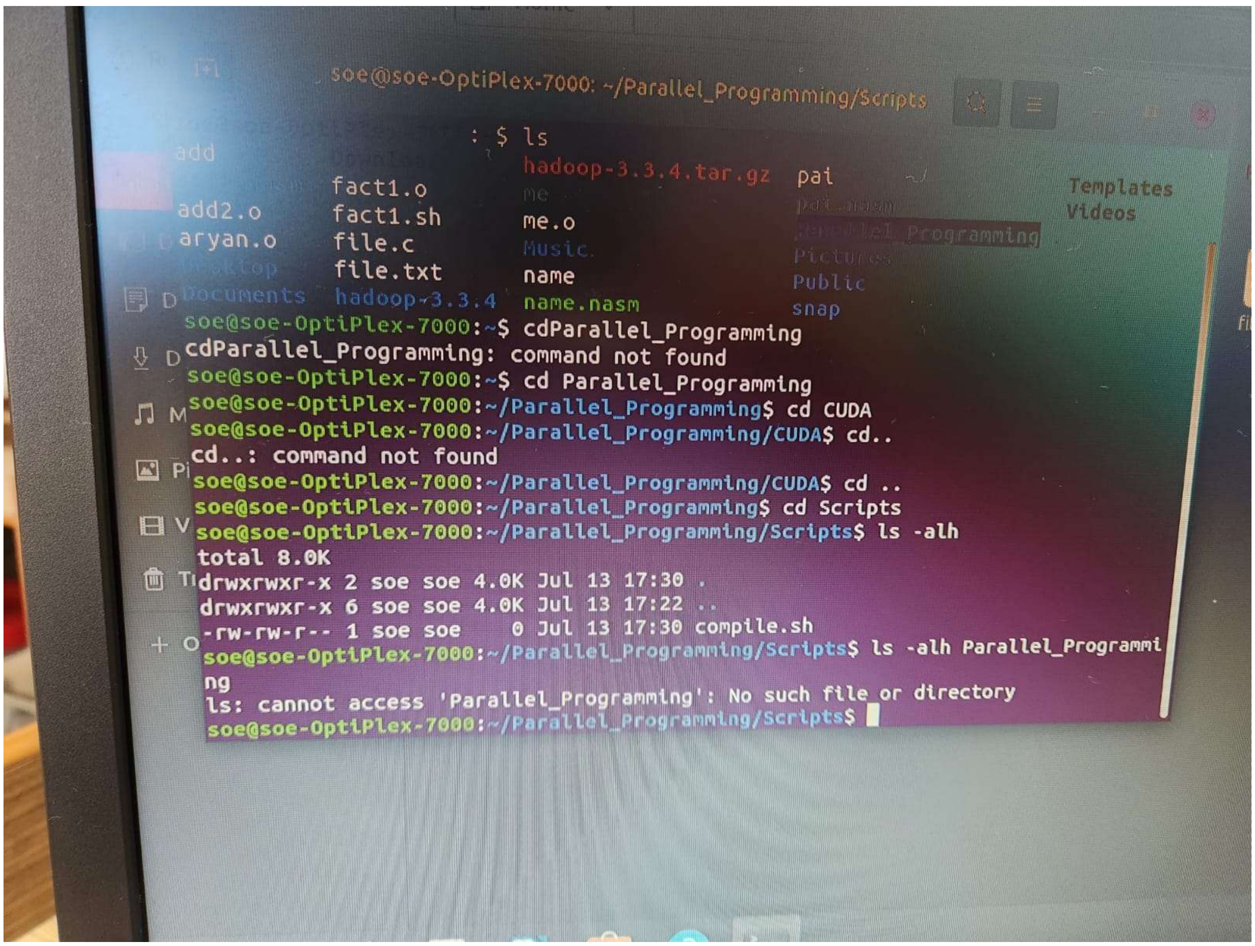

```
student@hpc-node01:~$ mkdir -p ~/Parallel_Programming/{OpenMP,MPI,CUDA,Scripts}
student@hpc-node01:~$ tree ~/Parallel_Programming/
/home/student/Parallel_Programming/
```

```
├── CUDA
├── MPI
├── OpenMP
└── Scripts

4 directories, 0 files
```

Q2. File Creation

Task: Create placeholder source files inside each respective directory (omp_hello.c, mpi_hello.c, vector_add.cu, compile.sh).




```
student@hpc-node01:~$ touch ~/Parallel_Programming/OpenMP/omp_hello.c
student@hpc-node01:~$ touch ~/Parallel_Programming/MPI/mpi_hello.c
student@hpc-node01:~$ touch ~/Parallel_Programming/CUDA/vector_add.cu
```

```
student@hpc-node01:~$ touch ~/Parallel_Programming/Scripts/compile.sh
student@hpc-node01:~$ tree ~/Parallel_Programming/
/home/student/Parallel_Programming/
├── CUDA
│   └── vector_add.cu
├── MPI
│   └── mpi_hello.c
├── OpenMP
│   └── omp_hello.c
└── Scripts
    └── compile.sh

4 directories, 4 files
```

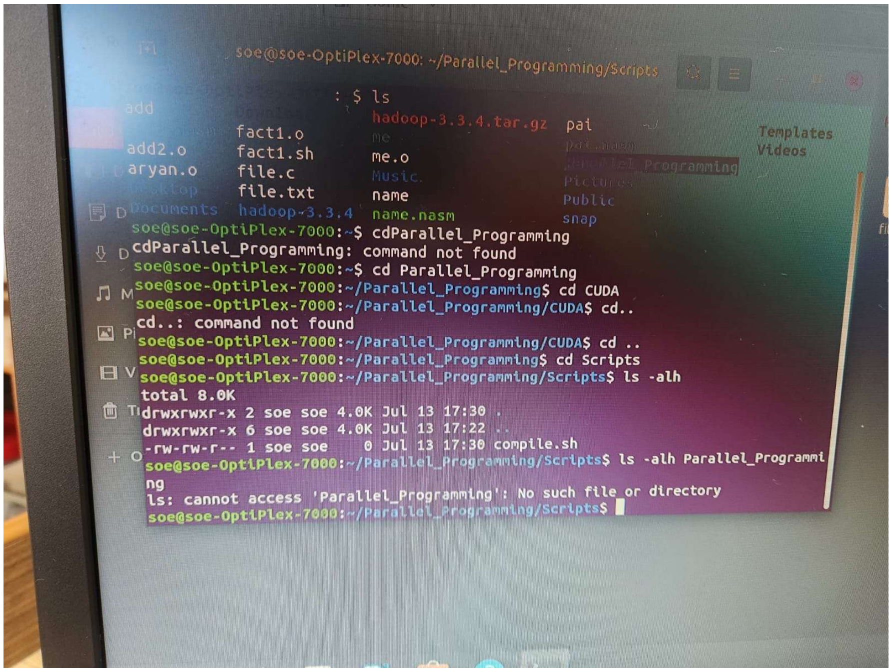
Q3. Navigation

Task: Navigate through directories without using the mouse: (a) move into CUDA directory, (b) return to home directory, (c) move to Scripts directory using a relative path.

```
student@hpc-node01:~$ cd ~/Parallel_Programming/CUDA
student@hpc-node01:~/Parallel_Programming/CUDA$ pwd
/home/student/Parallel_Programming/CUDA
student@hpc-node01:~/Parallel_Programming/CUDA$ cd ~
student@hpc-node01:~$ pwd
/home/student
student@hpc-node01:~$ cd Parallel_Programming/Scripts
student@hpc-node01:~/Parallel_Programming/Scripts$ pwd
/home/student/Parallel_Programming/Scripts
```

Q4. Listing Files

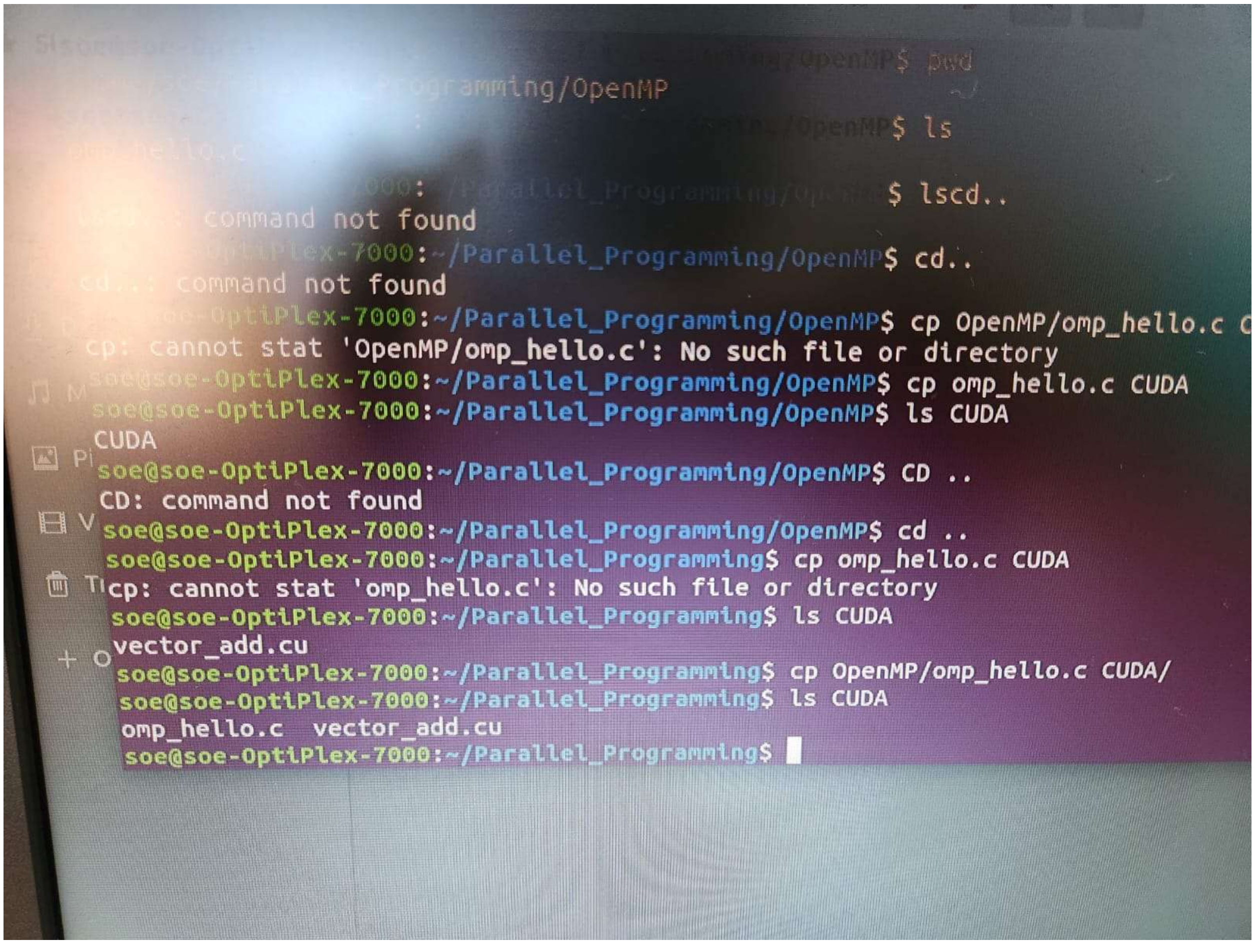
Task: Display all files, hidden files, and file sizes in human-readable format inside Parallel_Programming.




```
student@hpc-node01:~$ ls -la -h ~/Parallel_Programming/
total 24K
drwxr-xr-x 6 student student 4.0K Aug  3 10:15 .
drwxr-x--- 8 student student 4.0K Aug  3 10:15 ..
drwxr-xr-x 2 student student 4.0K Aug  3 10:15 CUDA
drwxr-xr-x 2 student student 4.0K Aug  3 10:15 MPI
drwxr-xr-x 2 student student 4.0K Aug  3 10:15 OpenMP
drwxr-xr-x 2 student student 4.0K Aug  3 10:15 Scripts
```

Question 5: Copying Files

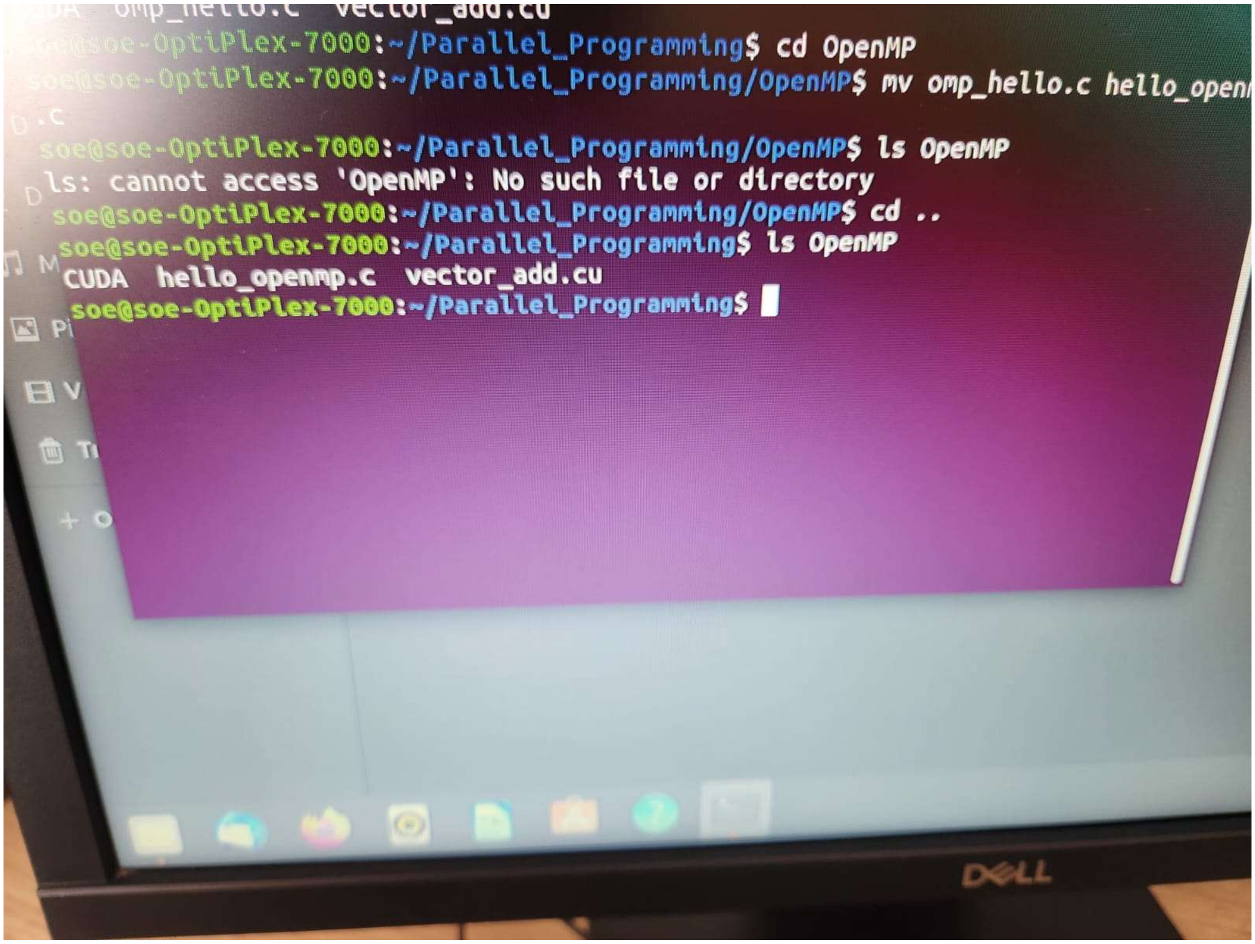
Task: Copy omp_hello.c from the OpenMP folder into the CUDA folder.




```
student@hpc-node01:~$ cp ~/Parallel_Programming/OpenMP/omp_hello.c
~/Parallel_Programming/CUDA/
student@hpc-node01:~$ ls -l ~/Parallel_Programming/CUDA/
total 0
-rw-r--r-- 1 student student 0 Aug  3 10:18 omp_hello.c
-rw-r--r-- 1 student student 0 Aug  3 10:15 vector_add.cu
```

Q6. Moving Files

Task: Move vector_add.cu from the CUDA folder into the OpenMP folder.




```
student@hpc-node01:~$ mv ~/Parallel_Programming/CUDA/vector_add.cu
~/Parallel_Programming/OpenMP/
student@hpc-node01:~$ ls -l ~/Parallel_Programming/OpenMP/
total 0
-rw-r--r-- 1 student student 0 Aug  3 10:15 omp_hello.c
-rw-r--r-- 1 student student 0 Aug  3 10:15 vector_add.cu
student@hpc-node01:~$ ls -l ~/Parallel_Programming/CUDA/
total 0
-rw-r--r-- 1 student student 0 Aug  3 10:18 omp_hello.c
```

Q7. Renaming Files

Task: Rename omp_hello.c inside the CUDA directory to hello_openmp.c.

```
student@hpc-node01:~$ mv ~/Parallel_Programming/CUDA/omp_hello.c
~/Parallel_Programming/CUDA/hello_openmp.c
```

```
student@hpc-node01:~$ ls -l ~/Parallel_Programming/CUDA/
total 0
-rw-r--r-- 1 student student 0 Aug  3 10:18 hello_openmp.c
```

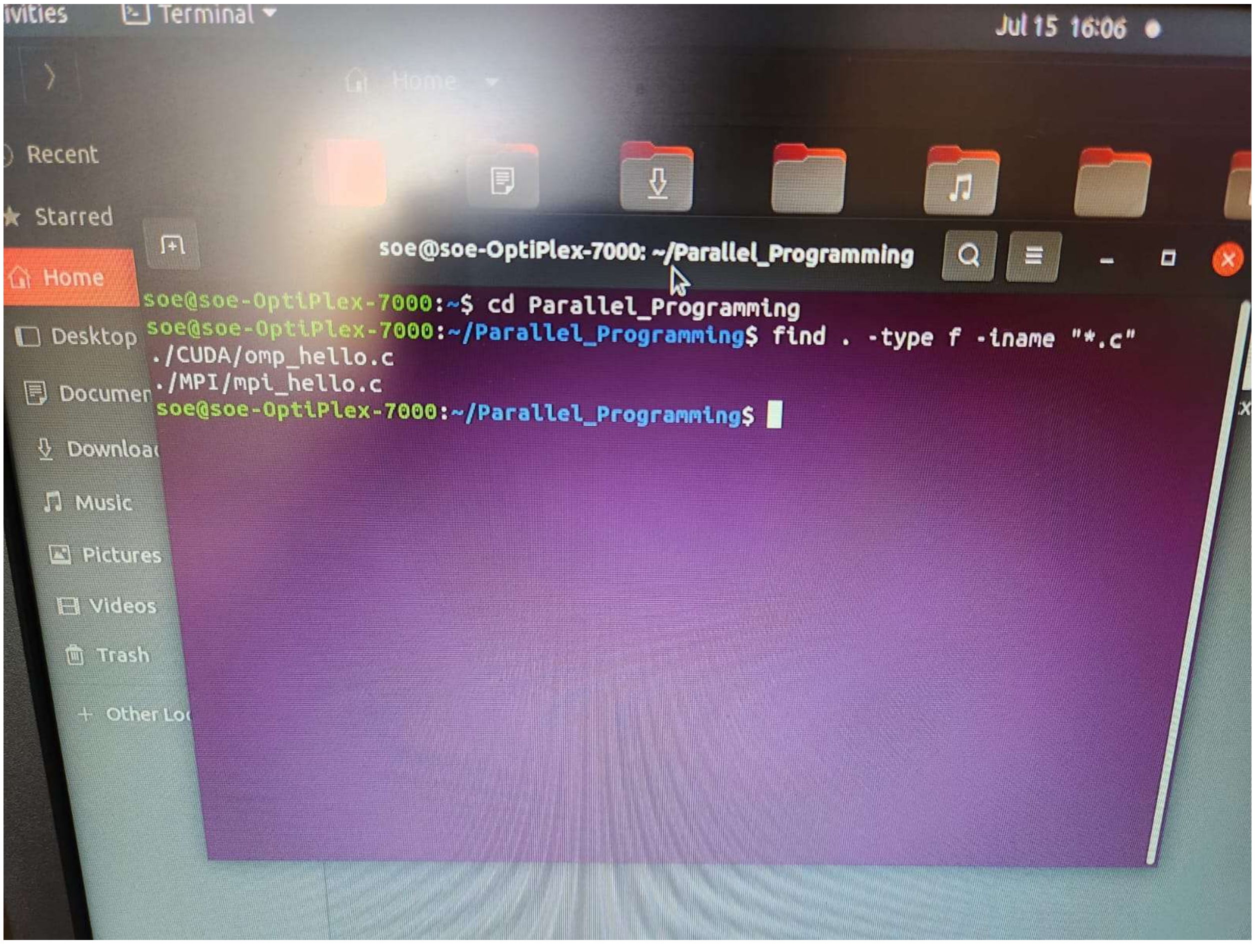
Q8. Deleting Files

Task: Delete hello_openmp.c without deleting the CUDA directory.

```
student@hpc-node01:~$ rm ~/Parallel_Programming/CUDA/hello_openmp.c
student@hpc-node01:~$ ls -la ~/Parallel_Programming/CUDA/
total 8
drwxr-xr-x 2 student student 4096 Aug  3 10:22 .
drwxr-xr-x 6 student student 4096 Aug  3 10:15 ..
```

Q9. Searching

Task: Find every file having the extension .c inside the Parallel_Programming directory.



```
student@hpc-node01:~$ find ~/Parallel_Programming/ -type f -name "*.c"
/home/student/Parallel_Programming/MPI/mpi_hello.c
/home/student/Parallel_Programming/OpenMP/omp_hello.c
```

Q10. Finding Current Location

Task: Display the current working directory.

```
student@hpc-node01:~/Parallel_Programming$ pwd
/home/student/Parallel_Programming
```

Q11. Viewing File Contents

Task: Create a text file containing five Linux commands and view its contents using cat, more, less, head, and tail.

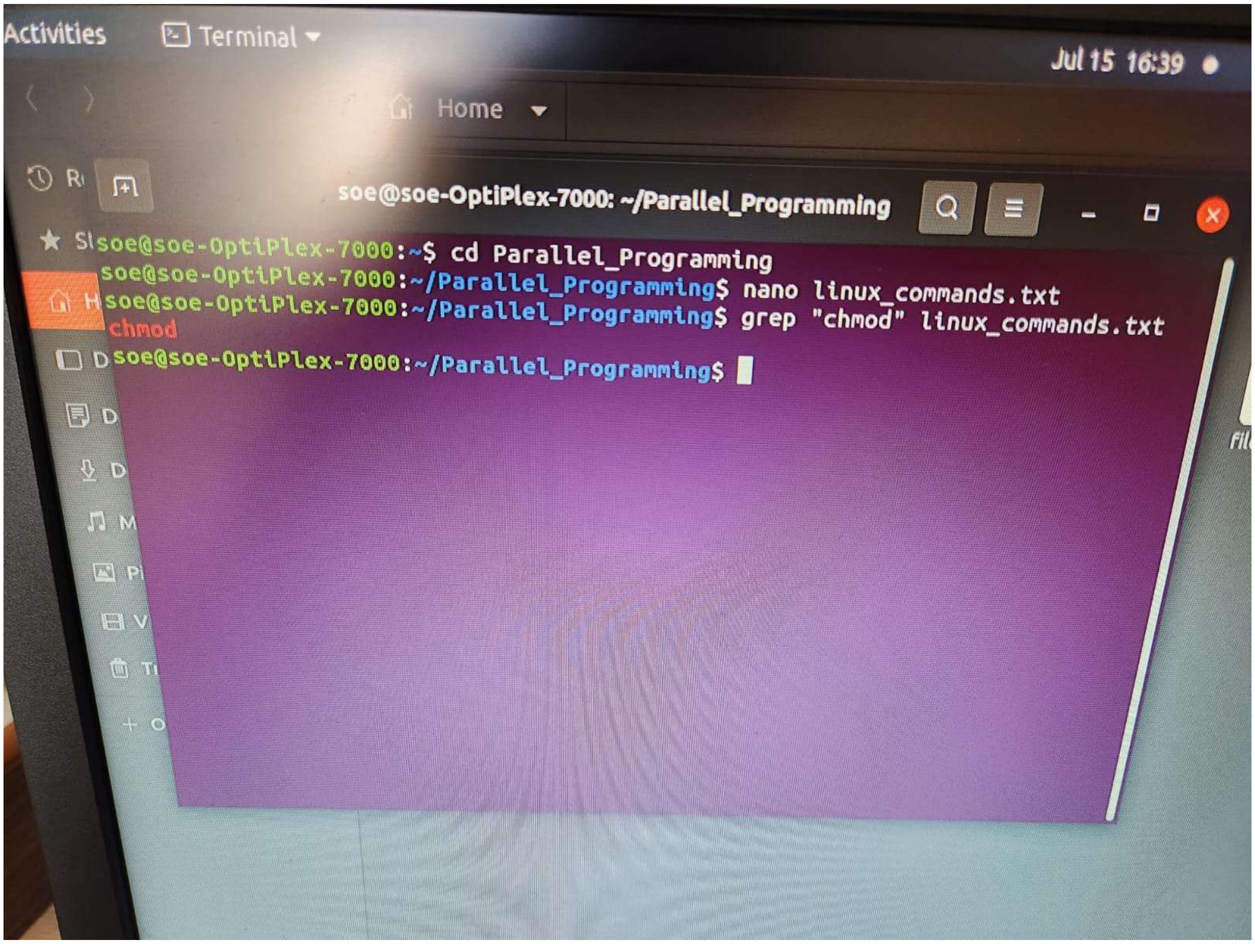
```
student@hpc-node01:~$ cat commands.txt
1. ls - Lists files and directories
2. cd - Changes current working directory
3. pwd - Displays absolute working path
4. mkdir - Creates new directory
5. chmod - Changes file access permissions

student@hpc-node01:~$ head -n 3 commands.txt
1. ls - Lists files and directories
2. cd - Changes current working directory
3. pwd - Displays absolute working path

student@hpc-node01:~$ tail -n 2 commands.txt
4. mkdir - Creates new directory
5. chmod - Changes file access permissions
```

Q12. Searching Text

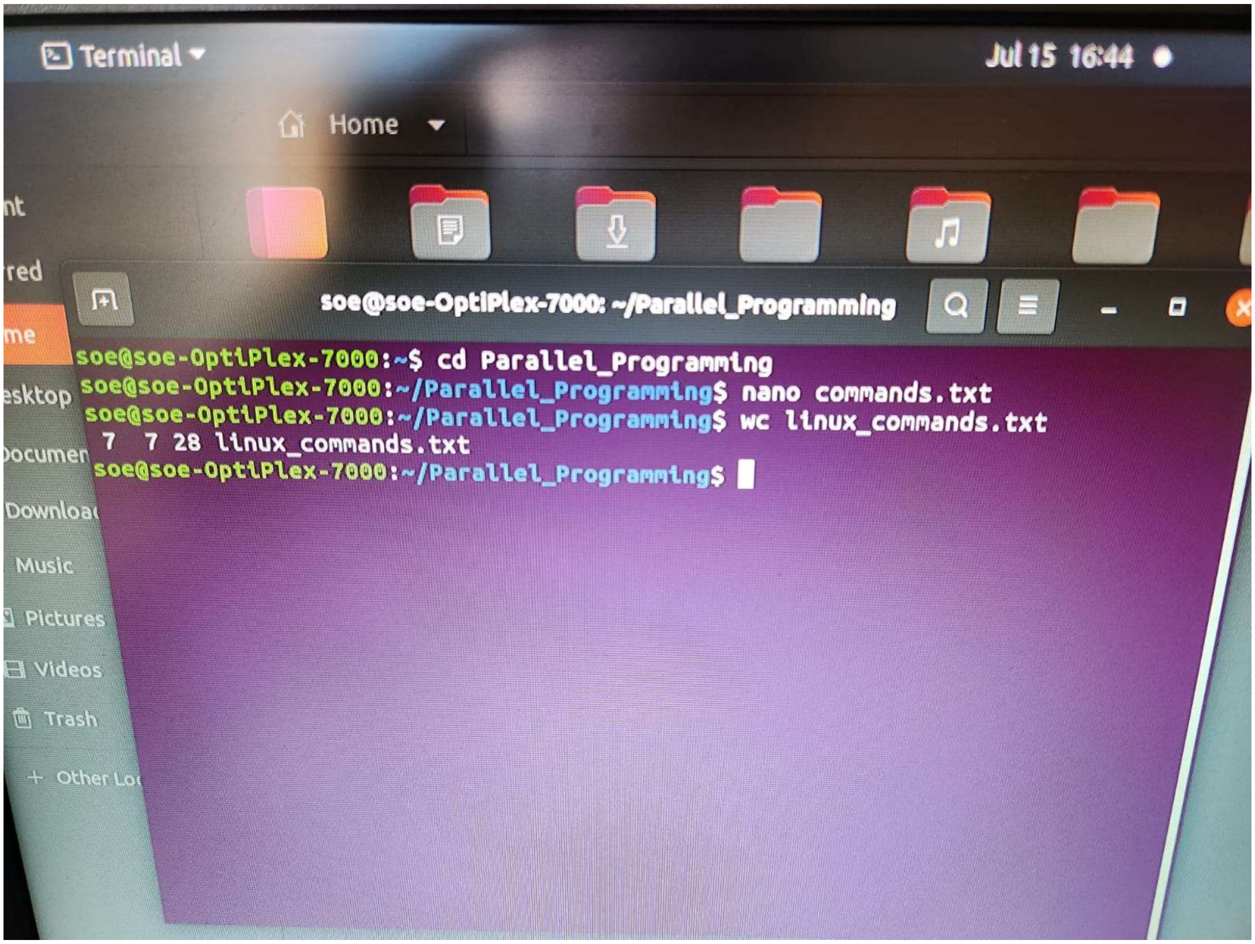
Task: Create a file containing Linux command names and search for 'chmod' using grep.




```
student@hpc-node01:~$ grep -n "chmod" cmd_list.txt
4:chmod
```

Q13. Counting

Task: Count the number of lines, words, and characters in a text file using wc.




```
student@hpc-node01:~$ wc commands.txt
  5  26 195 commands.txt

# Breakdown:
# Lines      : 5
# Words      : 26
# Characters : 195
```

Q14. File Permissions

Task: Make compile.sh executable using chmod and verify permission change using ls -l.

```
student@hpc-node01:~$ ls -l ~/Parallel_Programming/Scripts/compile.sh
-rw-r--r-- 1 student student 0 Aug  3 10:15
/home/student/Parallel_Programming/Scripts/compile.sh
```

```
student@hpc-node01:~$ chmod +x ~/Parallel_Programming/Scripts/compile.sh
student@hpc-node01:~$ ls -l ~/Parallel_Programming/Scripts/compile.sh
-rwxr-xr-x 1 student student 0 Aug  3 10:15
/home/student/Parallel_Programming/Scripts/compile.sh
```

Q15. Compression

Task: Compress Parallel_Programming into parallel.tar.gz and extract it into another directory.

```
student@hpc-node01:~$ tar -czvf parallel.tar.gz -C ~ Parallel_Programming
Parallel_Programming/
Parallel_Programming/CUDA/
Parallel_Programming/MPI/
Parallel_Programming/MPI/mpi_hello.c
Parallel_Programming/OpenMP/
Parallel_Programming/OpenMP/vector_add.cu
Parallel_Programming/OpenMP/omp_hello.c
Parallel_Programming/Scripts/
Parallel_Programming/Scripts/compile.sh

student@hpc-node01:~$ mkdir -p ~/Extracted_Backup && tar -xzvf parallel.tar.gz -C
~/Extracted_Backup/
Parallel_Programming/
Parallel_Programming/CUDA/
Parallel_Programming/MPI/
Parallel_Programming/MPI/mpi_hello.c
Parallel_Programming/OpenMP/
Parallel_Programming/OpenMP/vector_add.cu
Parallel_Programming/OpenMP/omp_hello.c
Parallel_Programming/Scripts/
Parallel_Programming/Scripts/compile.sh
```

Q16. Hello World Script

Task: Write a shell script (hello.sh) that prints 'Welcome to Parallel Programming'.

```
student@hpc-node01:~$ chmod +x hello.sh
student@hpc-node01:~$ ./hello.sh
Welcome to Parallel Programming
```

Q17. Variables

Task: Write a script (student_info.sh) that stores Name, Roll Number, and Branch in variables and prints them.

```
student@hpc-node01:~$ ./student_info.sh
-------------------------------------
STUDENT INFORMATION DETAILS
-------------------------------------
Name        : Alex Mercer
Roll Number : 21HPC1048
Branch      : Computer Science and Engineering (HPC)
-------------------------------------
```

Q18. User Input

Task: Write a script (prompt_compile.sh) that asks the user to enter 'Program Name' and prints 'Compiling <Program Name>'.

```
student@hpc-node01:~$ ./prompt_compile.sh
Enter Program Name: omp_matrix_mult.c
Compiling omp_matrix_mult.c
```

Q19. Arithmetic Operations

Task: Write a script (calc.sh) to input two numbers and display their Sum, Difference, Product, and Quotient.

```
student@hpc-node01:~$ ./calc.sh
Enter First Number: 28
Enter Second Number: 8
--- ARITHMETIC RESULTS ---
Sum        : 36
Difference : 20
Product    : 224
Quotient   : 3.50
```

Q20. Even / Odd Check

Task: Write a shell script (even_odd.sh) that checks whether an input number is even or odd.

```
student@hpc-node01:~$ ./even_odd.sh
Enter a number: 64
The number 64 is EVEN.

student@hpc-node01:~$ ./even_odd.sh
Enter a number: 37
The number 37 is ODD.
```

Q21. Largest Among Three Numbers

Task: Write a shell script (largest.sh) that finds the largest among three input numbers.

```
student@hpc-node01:~$ ./largest.sh
Enter First Number  : 45
Enter Second Number : 92
Enter Third Number  : 68
The largest number among 45, 92, and 68 is: 92
```

Q22. For Loop Implementation

Task: Write a shell script (for_loop.sh) that prints 'GPU Programming' 10 times using a for loop.

```
student@hpc-node01:~$ ./for_loop.sh
1: GPU Programming
2: GPU Programming
3: GPU Programming
4: GPU Programming
5: GPU Programming
6: GPU Programming
7: GPU Programming
8: GPU Programming
9: GPU Programming
10: GPU Programming
```

Q23. While Loop Implementation

Task: Write a shell script (while_loop.sh) that prints numbers from 1 to 20 using a while loop.

```
student@hpc-node01:~$ ./while_loop.sh
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
```

Q24. File Existence Check

Task: Write a script (check_file.sh) that checks whether a given file exists. Print 'File Found' if true, otherwise 'File Not Found'.

```
student@hpc-node01:~$ ./check_file.sh
Enter file path to check: Parallel_Programming/MPI/mpi_hello.c
File Found

student@hpc-node01:~$ ./check_file.sh
Enter file path to check: non_existent_file.txt
File Not Found
```

Q25. Counting Files in Directory

Task: Write a shell script (count_files.sh) that counts the number of files present in the current directory.

```
student@hpc-node01:~$ ./count_files.sh
Number of files in current directory: 8
```

Q26. OpenMP Compilation Script

Task: Write a shell script (compile_openmp.sh) that compiles an OpenMP program using gcc with -fopenmp flag, checks whether compilation succeeded, and displays an appropriate status message.

```
student@hpc-node01:~$ ./compile_openmp.sh
Compiling OpenMP program: Parallel_Programming/OpenMP/omp_hello.c ...
[SUCCESS] Compilation successful! Executable generated at:
Parallel_Programming/OpenMP/omp_hello
```

Q27. Batch Execution with Thread Counts

Task: Write a shell script (batch_run.sh) that runs the OpenMP executable for thread counts 1, 2, 4, and 8, displaying execution time for each run.

```
student@hpc-node01:~$ ./batch_run.sh
==================================================
  HPC BATCH EXECUTION: OPENMP THREAD SCALING
==================================================

--> Running with OMP_NUM_THREADS = 1
    Execution Time: 0.00241 seconds

--> Running with OMP_NUM_THREADS = 2
    Execution Time: 0.00185 seconds

--> Running with OMP_NUM_THREADS = 4
    Execution Time: 0.00122 seconds

--> Running with OMP_NUM_THREADS = 8
    Execution Time: 0.00098 seconds

==================================================
```

Q28. Command Line Arguments

Task: Write a shell script (add_args.sh) that accepts two numbers as command-line arguments ($1, $2) and prints their sum.

```
student@hpc-node01:~$ ./add_args.sh 150 350
Sum of 150 and 350 is: 500

student@hpc-node01:~$ ./add_args.sh 50
Usage: ./add_args.sh <num1> <num2>
```

Q29. Backup Script

Task: Write a shell script (backup_c.sh) that creates a backup of all .c files into a folder named Backup.

```
student@hpc-node01:~$ ./backup_c.sh
Locating and backing up all .c files into 'Backup' ...
[SUCCESS] Backup completed. Contents of 'Backup':
total 16
drwxr-xr-x 2 student student 4096 Aug  3 10:45 .
```

```
drwxr-xr-x 7 student student 4096 Aug  3 10:45 ..
-rw-r--r-- 1 student student  215 Aug  3 10:45 mpi_hello.c
-rw-r--r-- 1 student student  180 Aug  3 10:45 omp_hello.c
```

Q30. Mini Project: Automated HPC Environment Setup

Task: Create a shell script called project_setup.sh that automatically:

1. Creates folders: OpenMP, MPI, CUDA, and Results
2. Creates placeholder source files in each folder
3. Gives execute permission (+x) to all .sh files
4. Displays the created directory structure
5. Prints 'Project setup completed successfully.'

```
student@hpc-node01:~$ ./project_setup.sh
Starting HPC Project Setup...
-------------------------------------
CREATED DIRECTORY STRUCTURE:
-------------------------------------
OpenMP
└── omp_main.c
MPI
└── mpi_main.c
CUDA
└── cuda_main.cu
Results
└── benchmark_results.txt

0 directories, 4 files
-------------------------------------
Project setup completed successfully.
```